Everything Is Relative

How quantum physics stands to relativity: on the postulate that whatever can exist keeps the one light cone, the leading quantum correction to gravity's second-order law, on its string effective-action reading, is held to zero

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Abstract

We prove one statement and mark where it stops. Whatever can exist carries energy and momentum. It is then subject to relativity in three senses, each stronger than the last:

1. its invariants are Lorentz invariants, so it moves on or inside the light cone (*special*);
2. it couples to one metric with one strength, so everything falls alike and everything gravitates (*general*);
3. it keeps the one light cone. Every front of every field, gravity's own included, lies on or inside the null cone of that metric, at every point of every background that can exist (*one cone*).

Senses (1) and (2) are proved kind by kind. They use exact computation and the theorems of Lovelock, Deser, and Peres and Terno, and they are tested to the precision of GW170817, MICROSCOPE and Cassini. Sense (3) is a postulate, the one-cone principle. It is stated as a postulate and never derived.

The relation to quantum physics runs through three results:

- **Quantum information lives inside the causal structure.** Measurement cannot violate relativistic causality, and no local operation on one half of an entangled pair changes what the other half can observe. We prove the second fact in general and check it by exact computation on every two-qubit pure state.
- **The one constant classical relativity leaves open is the leading quantum correction.** Lovelock's theorem leaves gravity's second-order law exactly Einstein's in four dimensions. In five it admits one more coupling, the Gauss–Bonnet α, which the string effective action reads as "the leading order quantum correction". We compute, from the field equations, that with $\alpha \neq 0$ gravity's own waves leave the light cone, for either sign of α within the range where the theory has a vacuum. On a black hole the three cones fan out about the light cone, constrained by the identity $2(c_T - 1) + 2(c_V - 1) + (c_S - 1) = 0$, and they coincide with it only when $\alpha m = 0$. On a plane gravitational wave the displacements likewise sum to zero. The four-dimensional route by which the same term acts, a scalar coupled to the Gauss–Bonnet density, fails the same test for every coupling with $f' \neq 0$ on backgrounds with $V = V' = 0$. Under the postulate, $\alpha = 0$ within that range.

- **Quantum corrections can be neglected only where one dimensionless number is small.** Buckingham's theorem on $\{G_5, \hbar, c, \ell, R\}$ leaves two numbers, and in the basis $\{l_5^2 R, \ell^2 R\}$ only $x = l_5^2 R$ carries $\hbar$. Under four named premises, and the postulate through Theorem 2, the two-derivative law holds to corrections that vanish as $x \to 0$, and is no longer guaranteed where x reaches one.

The limits are part of the result:

- the experiments bound departures only to their precision;
- the cone splitting is computed on named families of backgrounds, and is generic by Reall, Tanahashi and Way;
- the scattering argument of Camanho, Edelstein, Maldacena and Zhiboedov corroborates only for $|\alpha| \gg l_5^2$;
- theories with field equations beyond second order (Einsteinian cubic gravity, f(R)) are not tested;
- nothing here is a theory of quantum gravity.

What is proved is compatibility, a single causal structure that quantum physics and gravity share, and not unification.

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1 · Introduction

1.1 The question

Quantum physics and relativity are usually said to be incompatible. The sentence means several different things. One reading is that quantum measurement seems to act instantaneously at a distance. Another is that gravity resists quantisation as an ordinary field theory. A third is that the classical field equations must fail at high curvature, and no one knows the law that replaces them.

This paper poses the question differently:

> Everything that exists, with or without mass, is subject to relativity.

Asked this way, the question is not whether one theory can be derived from the other. It asks whether there is anything that can exist and is not subject to relativity, and whether quantum physics supplies such a thing.

1.2 What is proved, and what is postulated

We prove the statement in senses (1) and (2) of the abstract, kind by kind (§§4–8, Theorem 1). We then ask what relativity still leaves free. Beyond the cosmological constant, the answer is one constant: the Gauss–Bonnet coupling α of the five-dimensional second-order law. On the string effective-action reading of Charmousis and Dufaux, that constant is where the leading quantum correction enters gravity's second-order law.

Under the postulate that everything keeps the one light cone, we show that constant is zero (§10, Theorem 2). We also show that its four-dimensional scalar realisation is excluded, on backgrounds with $V = V' = 0$ (§11). Last, we derive where the classical law stops being guaranteed: the boundary at which quantum corrections can no longer be neglected (§12).

The postulate is named as such wherever it is used. Without it, Theorem 1 still stands. Theorem 2 then becomes a conditional: if $\alpha \neq 0$ within its existence range, gravity has fronts outside the light cone.

1.3 One statement, proved in steps

The paper proves one statement, in steps, through a chain of lemmas. Lemmas 1–5 give Theorem 1 (the statement in senses (1)–(2)). Lemma 6 with the postulate gives Theorem 2 ($\alpha = 0$); Lemma 7 with the postulate gives its four-dimensional corollary. Proposition 8 gives the boundary of the classical law. Sense (3) itself is postulated, not proved.

1.4 Scope

The paper is built from first principles and established results only. Each step is either computed here, deduced here, or taken from a published theorem or measurement with its hypotheses stated. Nothing else is needed to follow it.

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2 · Postulates, definitions, premises

Definition 1 (a kind). A *kind* is anything that can exist and acts on or carries anything else. It is represented by a symmetric stress–energy tensor T_{ab}, or carried by something that has one. Its *mass* m is defined by $p_a p^a = -m^2c^2$ where it has a four-momentum. *With or without mass* means $m \geq 0$, with $m = 0$ admitted.

Definition 2 (subject to relativity). A kind is

- *subject to special relativity* if its invariants are Lorentz invariants, so that it moves on ($m = 0$) or inside ($m > 0$) the light cone in every inertial frame;
- *subject to general relativity* if it couples to a single metric g with a universal strength, both falling and gravitating;
- *one-cone* if every characteristic front of its field equations, on every background, is on or inside the null cone of g. A front is *characteristic* when its normal ξ makes the principal symbol of the field equations degenerate on physical polarisations (Reall, Tanahashi and Way, §2.2, READ).

Postulate O (the one-cone principle). Everything that can exist keeps the one light cone: every front of every field, gravity's own included, at every point, on every background that can exist, lies on or inside the null cone of the one metric.

It strengthens what Lemmas 2–3 establish. Local Lorentz invariance and universal coupling fix one cone for signals in flat space and in the weak field. Postulate O asks the

same of the characteristic fronts on every background. It is not derived anywhere below, and every result that uses it says so.

Premises of the outside theorems. These are used only where named:

- **(PL) Lovelock.** The field equations of the metric are given by a symmetric, divergence-free, natural (0,2)-tensor depending on the metric and its first and second derivatives (locality, second order).
- **(PD) Deser.** A massless spin-2 field coupled to a conserved source must, for consistency, couple to the total stress tensor, its own included.
- **(PT) Peres and Terno.** Lorentz transformations are implemented in quantum theory by unitary transformations.
- **(EEP)** The Einstein equivalence principle, as tested.

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3 · The method

The proof is kind by kind. Every statement carries one of four labels:

- **computed**: an exact symbolic computation (sympy), re-run for this paper and asserted by its check (Appendix C), and, where a failure control is listed, shown to reject it;
- **deduced**: follows from the statements it cites;
- **READ**: quoted verbatim from the source, re-read at source for this paper, with page and equation, and used only on that source's hypotheses;
- **postulate**: assumed, named, and never derived.

Each lemma carries its own evidence, of one of three types:

- an exact computation, done in sympy and re-run for this paper (module named in Appendix A);
- a theorem from the literature, applied only on its hypotheses (READ, quoted);
- an experiment, quoted with its bound (READ), which bounds a departure and never fixes a value.

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4 · What can exist (Lemma 1)

Lemma 1. Every kind the statement concerns carries energy and momentum:

- matter, massive or massless: particles, scalar fields, gauge fields;
- vacuum energy;
- the tension of an extended object (a sheet);
- geometry itself (gravitational waves);
- information, through its carrier.

Evidence.

- Matter, vacuum and tension each have a stress tensor. A vacuum has T_ab = −ρ g_ab, and a sheet of tension τ has T_ab = −τ h_ab on its world-volume, with h its induced metric (deduced).
- For particles, `E^2 = p^2 c^2 + m^2 c^4`, which at m = 0 is E = pc (computed).
- Information is not a separate substance. Peres and Terno (quant-ph/0212023, §I.A, p. 1, READ): signal speeds are bounded "because information needs a material carrier, and the latter must obey the laws of physics. Information is physical (Landauer, 1991)."
- Again (§III.C, p. 13, READ): "information has to be carried by material objects, quantized or not."

Scope. Something with no energy and momentum at all would act on nothing and carry nothing. It is outside the statement and is not claimed.

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5 · Special relativity for every kind (Lemma 2)

Lemma 2. Let L be a Lorentz boost of rapidity y. Then:

- $L^T \eta L = \eta$;
- a massless four-momentum stays null and future-pointing (its energy becomes E e^y > 0);
- a massive one keeps p·p = $-\mu^2$ and its speed stays below c ($1 - v'^2 = \mu^2/E'^2$);
- a scalar's Lagrangian −½ ∂φ·∂φ − V, the gauge invariants F_ab F^ab and ε^abcd F_ab F_cd, a vacuum's stress −ρη_ab, and a sheet's Nambu–Goto density √(−det h) (for boosts along the sheet) are each unchanged.

All of this is computed, exact, for every y.

Test (READ). GW170817 with GRB 170817A (LIGO–Virgo, Fermi-GBM, INTEGRAL, arXiv:1710.05834, abstract and eq. (1), pp. 1 and 6) constrain "the difference between the speed of gravity and the speed of light to be between -3×10^{-15} and $+7 \times 10^{-16}$ times the speed of light". Will (arXiv:1403.7377, abstract, READ): the Einstein equivalence principle "is well supported by experiments such as the Eötvös experiment, tests of local Lorentz invariance and clock experiments."

What the test does and does not show. The bound is on one speed over a near-flat path. It says nothing about the characteristic cones of a strongly curved background (§9, qualification Q1).

Failure control. Replacing the Lorentz boost by a Galilean one makes the null-cone and metric checks fail.

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6 · The universality of gravitation (Lemma 3)

Lemma 3. Gravitation couples to every kind through one metric with one strength, with mass or without.

Evidence.

- **Metric theories (READ, Will, arXiv:1403.7377, §2.1, p. 6).** “it is possible to argue convincingly that if EEP is valid, then gravitation must be a”curved spacetime” phenomenon”, and “the only theories of gravity that can fully embody EEP are those that satisfy the postulates of”metric theories of gravity”“. Will calls this an argument, not a theorem, and it is used here as such.
- **Universal coupling from consistency (READ, Deser, gr-qc/0411023, *Gen. Rel. Grav.* 1 (1970) 9, p. 4).** The massless spin-2 field, coupled to a source, “must actually be coupled to the total stress-tensor including that of the φ-field itself” (p. 1), and “Consistency has therefore led us to universal coupling, which implies the equivalence principle.” This holds under (PD). Deser notes that consistency and linearity “may also be reconciled if locality is abandoned” (p. 1, footnote 2), so locality is a premise.
- **Geodesics carry no mass (computed).** For any static spherical metric, the Euler–Lagrange equations of ½ m g_ab ẋ^a ẋ^b are m times those of ½ g_ab ẋ^a ẋ^b, so the paths are independent of m. Light follows the same law with g_ab ẋ^a ẋ^b = 0. A force that knows the body’s mass makes the check fail (mutant).
- **Light bending (computed).** Light is bent by (1 + γ) 2GM/(c²b). This is 1.751 arcsec at the Sun’s limb with γ = 1.

Tests (READ).

- MICROSCOPE (arXiv:2209.15487, eq. (5), p. 5): η(Ti, Pt) = [−1.5 ± 2.3 (stat) ± 1.5 (syst)] × 10^{-15}.
- Cassini, as quoted in arXiv:1710.05834, p. 7: γ_EM − 1 = (2.1 ± 2.3) × 10^{-5}, “from the measurement of the Shapiro delay (at radio wavelengths) with the Cassini spacecraft (Bertotti et al. 2003)”.
- Gravity’s own waves feel the same Shapiro delay as light (arXiv:1710.05834, eq. (4), p. 7): $-2.6 \times 10^{-7} \leq \gamma_{GW} - \gamma_{EM} \leq 1.2 \times 10^{-6}$.

So the massless are gravitated as the massive are, and so is gravity itself, to these precisions.

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7 · The law of the one metric (Lemma 4)

Lemma 4 (Lovelock’s theorem, READ; the count computed). Under (PL), the symmetric, divergence-free, second-order natural (0,2)-tensors of a metric in n dimensions are spanned by the Lovelock tensors L_0, …, L_p, with p = ⌊(n − 1)/2⌋. In Navarro and Navarro’s statement (arXiv:1005.2386, Theorem 2.6, p. 9, READ):

> “Let (X, g) be a pseudo-riemannian manifold of dimension n and let p be the integer part of (n − 1)/2. The Lovelock tensors: L_0, . . . , L_p are a basis for the R-vector space of second-order natural (0,2)-tensors on X that are symmetric and divergence-free.”

Here L_0 is the metric and L_1 is Einstein's tensor (p. 9). The count gives $p(4) = 1$ and $p(5) = 2$ (computed), so:

- **In four dimensions** the only field equations are Einstein's with a cosmological constant.
- **In five** there is exactly one more term, Lanczos's tensor. It is "quadratic on the second derivatives of the metric, but only exists if X is of dimension strictly greater than 4" (Navarro and Navarro, p. 1, READ).

The theorem is Lovelock's (1970–72). Lanczos found the quadratic tensor in 1938. Both are named through Navarro and Navarro's short proof, which is the text re-read here; the original papers were not re-read.

The one constant, and why it is the quantum correction. Charmousis and Dufaux (hep-th/0202107) write the five-dimensional second-order action as $R + 12k^2 + \alpha(R_abcd\ R\^abcd - 4R_ab\ R\^ab + R^2)$ (eq. (2), p. 4), with $\Lambda = -6k^2$. They say it "is the most general action that will yield second order partial differential equations with respect to the metric components in 5 dimensions" (p. 5, READ). Of α they write (p. 3, READ):

> "Furthermore in an effective action approach of string theory, the Gauss-Bonnet term corresponds to the leading order quantum correction to gravity in particular in the case of the heterotic string [15]."

They cite Gross and Sloan, 1987, not re-read here. Their conclusion repeats it: the coupling "provides a window to the leading quantum gravity correction coming from string theory" (§V, p. 16, READ).

The Gauss–Bonnet term is the Euler density of four dimensions. It "yields in $D = 4$ a boundary term hence a topological and not dynamical contribution" (p. 3, READ). So:

- in four dimensions the term acts only if something varies its coefficient, a scalar $f(\varphi)$ (§11);
- in five it acts directly through α (§10).

Qualification carried. The phrase "leading order quantum correction" is Charmousis and Dufaux's reading inside the string effective action. This paper uses it as their identification and does not derive it. Lovelock's theorem fixes the law up to α. Deser's consistency argument does not require α (it builds Einstein's equations). So the one freedom relativity leaves gravity's second-order law is the one place a quantum correction can enter at second order, on Charmousis and Dufaux's string effective-action reading. Corrections beyond second order lie outside (PL) (§12).

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8 · Information and entanglement (Lemma 5)

Lemma 5. Quantum measurement, and every local operation on one part of an entangled system, transmits nothing outside the light cone.

READ. Peres and Terno (quant-ph/0212023, §III.C, p. 13):

> “Therefore quantum measurements do not allow any information to be transmitted faster than the characteristic velocity that appears in the Green’s functions of the particles emitted in the experiment. In a Lorentz invariant theory, this limit is the velocity of light. In summary, relativistic causality cannot be violated by quantum measurements. The only physical assumption that is needed to prove this assertion is that Lorentz transformations of the spacetime coordinates are implemented in quantum theory by unitary transformations of the various operators.”

Deduced, in general. Let Alice act on her system A with any operation, a unitary or a measurement whose outcome is not sent to Bob, written with Kraus operators $K_j \otimes 1$ with $\Sigma_j K_j^\dagger K_j = 1$. Bob’s reduced state after it is $\mathrm{Tr}_A[\Sigma_j (K_j \otimes 1)\, \rho\, (K_j^\dagger \otimes 1)] = \mathrm{Tr}_A[(\Sigma_j K_j^\dagger K_j \otimes 1)\, \rho] = \mathrm{Tr}_A \rho$, by cyclicity of the partial trace over A. So nothing Alice does locally changes any statistic Bob can measure. This is the no-communication theorem (Peres and Terno, §II.E; the section was not re-read here, and the one-line proof above is given in full instead).

Computed (this paper), as a check of that proof. Take an arbitrary two-qubit pure state $|\psi\rangle = \Sigma\, a_i\, |i\rangle$, with four symbolic complex coefficients. Let Alice either apply a local unitary $U(\theta, \varphi) \otimes 1$ from a two-parameter family, or make a non-selective projective measurement in the basis $U|0\rangle$, $U|1\rangle$. In both cases Bob’s reduced density matrix ρ_B is unchanged, identically in a_i, θ, φ. A non-local gate (CNOT) passed off as Alice’s local operation does change ρ_B, and the check catches it (mutant `signal`). Entanglement therefore correlates without signalling. Whatever it carries, it carries through the causal structure of Lemmas 2–3.

What is not claimed. Peres and Terno also show that “Quantum entropy is not a Lorentz covariant concept” (abstract, READ). The relativity of information is that of its carrier, not an invariance of every information-theoretic quantity.

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9 · The statement in senses (1)–(2) (Theorem 1)

Theorem 1. Every kind that can exist, with mass or without, satisfies the following:

1. It carries energy and momentum (Lemma 1).
2. Its invariants are Lorentz invariants. It moves on the light cone if massless and inside it if massive, in every inertial frame (Lemma 2).
3. It couples to one metric with one strength, falling alike and gravitating (Lemma 3).
4. It carries information only through its carrier and signals nothing outside the light cone (Lemma 5).
5. The one metric obeys Einstein’s law in four dimensions, and Einstein’s law with at most one constant more, α, in five (Lemma 4).

The theorem is deduced from Lemmas 1–5. It is exact on the premises of the outside theorems and of Will’s argument, and tested to the experiments’ precision.

Scope (qualification Q1, §13). What Theorem 1 proves of "keeping the light cone" is special-relativistic. It covers flat-space boost invariance, together with GW170817's near-flat equality of speeds. It proves nothing about the characteristic cones of a curved background. That stronger statement is Postulate O, and only Theorem 2 uses it.

What Theorem 1 leaves open. Its fifth clause allows one constant, α. If α is nonzero, gravity's fronts leave the light cone (§10), so under Postulate O the constant is fixed (Theorem 2).

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10 · The coupling evaluated by the one light cone (Lemma 6, Theorem 2)

10.1 The computation

In five dimensions, write the field equations as $E = G + \Lambda + \alpha H$. The Einstein and Gauss–Bonnet tensors are written with the generalised Kronecker delta in an orthonormal frame: $G = -(1/4)\,\delta^{(3)}\,R$, $H = -(1/8)\,\delta^{(5)}\,R\,R$.

The normalisation is checked before use, against two exact solutions:

- **Tangherlini.** The five-dimensional Schwarzschild–Tangherlini metric $f = 1 - m/r^2$ comes out Ricci-flat, with Kretschmann scalar $72\,m^2/r^8$.
- **Boulware–Deser–Cai.** The hole $f = \kappa + r^2(1 - q)/(4\alpha)$, with $q^2 = 1 + 8\alpha m/r^4 + 4\alpha\Lambda/3$, solves $E = 0$ exactly for $\kappa = 1$ and $\kappa = 0$, with α, m, Λ and r left symbolic. The solution is Boulware and Deser's (1985) and Cai's (2002); it is quoted as Reall, Tanahashi and Way's eq. (5.15), READ, and Charmousis and Dufaux's eq. (17), READ.

The principal symbol, the part of the linearised equations that decides where waves can go, is

$$P(\xi)\cdot t = -\tfrac{1}{2}\,[\delta^{(3)}\,\xi\xi\,t + \alpha\,\delta^{(5)}\,\xi\xi\,t\,R].$$

It is computed to be gauge invariant (it kills $\xi X + X\xi$), divergence-free ($\xi\cdot P = 0$) and symmetric. A front with normal ξ is characteristic when P(ξ) has a non-gauge kernel. On transverse tensors this is the vanishing of a determinant, a polynomial in ξ_0 (Reall, Tanahashi and Way, §§2.2 and 3.1, READ). Doubling the Gauss–Bonnet part of the symbol against the field equation is caught (mutant).

10.2 Lemma 6: the cones split from the light cone when α ≠ 0

(a) Einstein's law. At α = 0, every characteristic front is null (computed on the family of (b); on any background, READ: Reall, Tanahashi and Way, arXiv:1406.3379, §1, p. 2): "For the Einstein equation, a hypersurface is characteristic if, and only if, it is null:"gravity travels at the speed of light"."

(b) Any static pair-diagonal background, spherical or planar ($-f\,dt^2 + dr^2/f + r^2 d\Sigma^2$). Let the pair curvatures be $K_1 = R^{tr}_{tr}$, $K_2 = R^{ti}_{ti} = R^{ri}_{ri}$ and

$K_3 = R^{ij}_{ij}$. A front moving along the sphere is characteristic at $\xi_0^2 = c|\xi|^2$. The graviton's five polarisations fall on three cones:

- $c_1 = (1 + 4\alpha K_1)/(1 + 4\alpha K_2)$, twice;
- $c_2 = (1 + 4\alpha K_2)/(1 + 4\alpha K_3)$, twice;
- $c_3 = (3 + 32\alpha K_2 + 64\alpha^2 K_2^2 - 4\alpha(K_1 + K_3) - 16\alpha^2 K_1 K_3) / (3(1 + 4\alpha K_2)(1 + 4\alpha K_3))$, once.

Radially every front is null. Every c is 1 when $K_1 = K_2 = K_3$, that is on any maximally symmetric background (flat, de Sitter or anti-de Sitter), which has no Weyl curvature. All of this is computed, exact.

(c) On the Boulware–Deser hole, exactly. With $A = (1 + 4\alpha\Lambda/3)/q^2$, the cones are

$c_T = 3 - 2A$ (twice), $c_V = A$ (twice), $c_S = 2A - 1$ (once),

for the spherical ($\kappa = 1$) and the planar ($\kappa = 0$) hole alike. These are Reall, Tanahashi and Way's tensor, vector and scalar cones (their eqs. (5.13) and (5.17) at d = 5, pp. 24–25, READ), here obtained from the field equations directly. Two routes give one answer.

The three forms satisfy, identically in A,

$2(c_T - 1) + 2(c_V - 1) + (c_S - 1) = 0$,

so the cones fan out about the light cone and cannot all lie inside it unless all three equal it, which needs A = 1. And $A - 1 = -8\alpha m/(r^4 q^2)$ (computed), which is nonzero whenever $\alpha m \neq 0$. So for $m > 0$:

- for $\alpha m > 0$ the tensors are outside it;
- for $\alpha m < 0$ the vectors and the scalar are.

For the fastest cone, $c - 1 = 16|\alpha|m/(r^4 q^2)$ (computed; c is a squared speed, so the speed exceeds light by about $8|\alpha|m/(r^4 q^2)$ to first order), at every radius outside the horizon. Reall, Tanahashi and Way (p. 22, READ): "If $c_A(r) > 1$ then the outermost graviton cone lies outside the light cone: gravity travels faster than light. We will see below that this is often the case, including arbitrarily far from the black hole."

(d) On gravity's own wave. A plane gravitational wave (Weyl type N, profile $\mathrm{diag}(2, -1, -1)\varepsilon$) solves the Gauss–Bonnet law exactly (computed). Reall, Tanahashi and Way (§4.1, p. 13, READ): "It is easy to see that any such spacetime is a solution of Lovelock theory with $\Lambda = 0$."

A graviton crossing it is characteristic at $\xi^2 = -(l\cdot\xi)^2\,\omega$, with

$\omega = 8\alpha\varepsilon$ (once), $4\alpha\varepsilon$ (twice), $-8\alpha\varepsilon$ (twice),

which sum to zero (computed). These agree with Reall, Tanahashi and Way's $\omega_I = -32\, k_2\, \nu_I$ (eq. (4.11), p. 17, READ) with $k_2 = -\alpha/8$. Some ω is positive for either sign of α, which is a front outside the light cone. RTW (p. 14, READ): "there is generically at least one positive ω_I and so this outermost cone lies outside the light cone".

On an anti-de Sitter background of curvature $-k^2$ (every pair curvature equal to $-k^2$) the wave solves the law with $\Lambda = -6k^2(1 - 2\alpha k^2)$, and every ω is divided by $1 - 4\alpha k^2$ (computed). In Charmousis and Dufaux's parametrisation $\Lambda = -6k_CD^2$, on their stable branch, $1 - 4\alpha k^2 = \sqrt{(1 - 8\alpha k_CD^2)}$ (computed). This is positive inside the existence range and vanishes at its endpoint $8\alpha k_CD^2 = 1$, where the linearised graviton degenerates (qualification Q3).

10.3 Theorem 2

Theorem 2 (under Postulate O). In five dimensions, under (PL), and for every α for which the law has a vacuum, the Gauss–Bonnet coupling is zero: $\alpha = 0$, and gravity's second-order law is Einstein's exactly.

Proof (deduced). The five-dimensional law can carry gravity's own waves and holds black holes, whose Weyl curvature is not zero (Tangherlini's Kretschmann $72\ m^2/r^8 \neq 0$). Write $\Lambda = -6k_CD^2$ (either sign of Λ). The existence range is $1 - 8\alpha k_CD^2 \geq 0$, where the maximally symmetric vacuum exists; for anti-de Sitter, Charmousis and Dufaux write "the maximally symmetric solution is defined only for $1 \geq 8\alpha k^2$ (for $\alpha < 0$ there is no such restriction)" (p. 10, READ, their k^2 being k_CD^2), and outside it "no solutions exist" (p. 10, READ). Neither background of Lemma 6 exists outside that range, and Theorem 2 makes no claim about such α. Inside it, $\alpha \neq 0$ puts a polarisation of gravity outside the light cone, whatever α's sign or size: on the hole (Lemma 6 (c)) for $m > 0$, and on the wave (Lemma 6 (d)) away from the endpoint $8\alpha k_CD^2 = 1$. At that endpoint the wave's cones degenerate and the hole alone carries the proof ($A = 0$ there, so $c_T = 3$). Postulate O forbids any front outside the one light cone. At $\alpha = 0$ every front is null (Lemma 6 (a)). ∎

Corroboration, within its range. Camanho, Edelstein, Maldacena and Zhiboedov (arXiv:1407.5597, READ) give a scattering argument. A graviton crossing a shock wave in Gauss–Bonnet theory gets a time advance: "by choosing different polarizations we can get time advance for $b^2 \sim |\lambda_GB|$ for any sign of λ_GB" (§2.4, p. 12). So "the purely Gauss-Bonnet theory (2.17) is acausal" (p. 13). The violation "cannot be fixed by adding conventional particles with spins $J \leq 2$. But, it can be fixed by adding an infinite tower of extra massive particles with higher spins, $J > 2$" (abstract), and "a finite number of higher spin particles does not fix the problem" (§4.4, p. 33).

Their result holds only "For $\alpha \gg l_p^2$" (§1, p. 3). Of order-one coefficients they write: "then we have nothing to say". It is also a tree-level statement. It corroborates Theorem 2 only for $|\alpha| \gg l_5^2$ (qualification Q2), l_5 the five-dimensional Planck length of §12. Theorem 2 itself does not use it: Lemma 6 is local and pointwise, not an asymptotic-causality argument.

What Theorem 2 does not say.

- It does not say gravity has been seen to keep the light cone in five dimensions. The effect at $\alpha \neq 0$ near a hole, in the squared speed, is $16|\alpha|m/(r^4q^2)$, and no measurement reaches it. $\alpha = 0$ is deduced, not measured.
- It does not say no world could hold an infinite tower of higher spins. Such a tower would take the law outside (PL), which Lemma 4 used.

- The split is computed on the families named: static pair-diagonal backgrounds and type N waves. It is generic by Reall, Tanahashi and Way (abstract: “characteristic hypersurfaces are generically all non-null”). It is not proved at every point where the Weyl curvature is nonzero (qualification Q4).
- It is computed at d = 5. Reall, Tanahashi and Way’s statements hold for Lovelock theories in general d (READ), but this paper computes only d = 5.

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11 · The four-dimensional scalar–Gauss–Bonnet route excluded (Lemma 7)

In four dimensions the Gauss–Bonnet density is topological. The leading correction can act only through a coupling that varies, in the theory

$S = \int\sqrt{(-g)}\,[R/2 - (\nabla\varphi)^2/2 - V(\varphi) + f(\varphi)\,L_{GB}]$.

Here f is any function: exponential for the dilaton, linear for the shift-symmetric case. A constant f does nothing, so the theory’s distinct content needs $f' \neq 0$.

Field equations (computed, exact). Varying the action gives the metric equation $G^a_b + 2\,\delta^{acde}_{bghk}\,(\nabla_c \nabla^g f)\,R_{de}{}^{hk} = T^a_b$, and the scalar equation $\Box\varphi - V' + f'\,L_{GB} = 0$. They were checked against a cosmology’s reduced action: the Euler–Lagrange equations in N, a and φ are $-a^3$, $-3a^2$ and a^3 times the covariant ones, for every f. Reading the Gauss–Bonnet term’s factor 2 as 1 is caught (mutant).

A cosmology is conformally flat, so this check cannot see the Weyl part. The first verification (§13) closed that gap with a static-spherical reduction (coefficient −8 from both the tt and rr equations) and with the Bianchi identity at −8, which fails at −4 and −16 (qualification Q7).

The principal symbol (computed) of the coupled system (t, ψ) is gauge invariant and divergence-free. It is symmetric as an action’s must be: the scalar row’s coupling to t is −½ the metric row’s coupling to ψ. With $f' = 0$ every front is null.

Lemma 7. Let $f'(\varphi_0) \neq 0$ at a point where $V(\varphi_0) = V'(\varphi_0) = 0$. Then the four-dimensional scalar–Gauss–Bonnet theory has fronts outside the light cone.

(a) On gravity’s own wave. A plane gravitational wave (type N, profile $\mathrm{diag}(1, -1)\varepsilon$) with $\varphi = \varphi_0$ constant solves the theory exactly ($L_{GB} = 0$ on it) (computed). The purely gravitational polarisation keeps the light cone. The two graviton–scalar polarisations are characteristic at

$\xi_0^2 = |\xi|^2/(1 \mp 8\sqrt{2}\,f'\varepsilon)$,

so one lies outside the light cone for either sign of f' (computed). The symmetry about the light cone is the symmetry of ω, equivalently of $1/\xi_0^2$, not of ξ_0^2 (qualification Q8).

(b) In a weakly coupled cosmology. The tensor fronts run at $c_T^2 = (1 + 8\ddot{f})/(1 + 8H\dot{f})$, twice (computed). This is the null cone of Reall’s effective metric $C^{ab} = g^{ab} - \beta'\,\nabla^a\nabla^b\Phi - \beta''$

$\nabla^a\Phi\,\nabla^b\Phi$ (arXiv:2101.11623, eq. (91), p. 18, READ), here with $\beta = 8f$. Two routes give one answer.

At exact solution points ($V = 0$, linear f, $H = 1$, $\dot\varphi = \pm 12/5$, $f' = \mp 1/480$, with the Friedmann equation holding exactly and $8|H\dot f| = 1/25$), $c_T^2 = 537/461 > 1$ (computed). The two points are one point under the reflection $\varphi \to -\varphi$, $f' \to -f'$. The sign of $c_T^2 - 1 = 8(\ddot f - H\dot f)/(1 + 8H\dot f) = 76/461$ is set by $\ddot f$ and $\dot f$, with $8\ddot f = 1363/11525$ (computed; qualification Q6).

Corroboration (READ). Serra, Serra, Trincherini and Trombetta (arXiv:2205.08551) treat "a massless (shift-symmetric) scalar coupled to the Gauss-Bonnet (GB) invariant" (p. 1). The scalar–graviton mixing gives "a non-diagonal transition amplitude between graviton and scalar, such that one of the propagation eigenmodes travels faster than what is allowed by the causal structure of the asymptotic spacetime" (§2, p. 2). The delays are $\Delta t_\pm = 2r_s(\log(b_0/b) \pm \sqrt{2}|\hat\alpha|/b^2)$ (eq. (12), p. 5). They find the time advance avoidable only if the effective theory's cutoff obeys $|\hat\alpha| \lesssim \log(b_0\Lambda)/\Lambda^2$ (eq. (13)), that is, by new physics at that scale.

Corollary (under Postulate O). The four-dimensional scalar–Gauss–Bonnet theory is excluded wherever its backgrounds have $V(\varphi_0) = V'(\varphi_0) = 0$: every point when $V = 0$, the shift-symmetric and massless-dilaton cases. This holds whether its scalar is read as part of gravity or as matter, because matter must keep the cone too (Lemma 1, Postulate O).

Not shown. The case $V \neq 0$ at φ_0 (for example $V = m^2\varphi^2/2$ with $f = \lambda\varphi^2$) is not shown (qualification Q5). Theories with field equations beyond second order in four dimensions, such as Einsteinian cubic gravity and f(R), lie outside (PL); their cone tests are not run.

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12 · Where quantum corrections can no longer be neglected (Proposition 8)

Where the classical law stops being guaranteed is not assumed here. Proposition 8 derives it, under named premises.

The premises, each named so it can be refused:

- **P1.** The constants of the five-dimensional law of §§7 and 10 are G_5, $\hbar$ and c, with a classical $\Lambda_5 = -6/\ell^2$ (an anti-de Sitter radius ℓ) and no other scale.
- **P2.** The classical part is Einstein's law, exact at two derivatives (Theorems 1–2, so under Postulate O). Everything beyond it carries $\hbar$ and vanishes as $\hbar \to 0$.
- **P3 (decoupling).** Where the curvature is far above $1/\ell^2$, the corrections do not grow with ℓ.
- **P4 (no free number).** No pure number is chosen.

Proposition 8. Under P1–P4 (and so under Postulate O, through P2), classical general relativity holds up to corrections that vanish as $x \to 0$, so it is accurate where $x \ll 1$ (for $\ell \gg l_5$, so that $\ell^2 R \gg 1$ there), and it is no longer guaranteed where x reaches order one, with $x = l_5^2\sqrt{K}$, $l_5^3 = G_5\hbar/c^3$, and K the Kretschmann scalar (the curvature's scale). The boundary is $x = 1$.

Proof.

1. **Two groups (computed).** By Buckingham's theorem, the dimension matrix of {G_5, ħ, c, ℓ, R} in (mass, length, time) has rank 3. So exactly two independent dimensionless numbers exist: $l_5^2 R$ and $\ell^2 R$.
2. **In this basis, only one carries ħ (computed).** $l_5^2 R$ carries ħ (as ħ^{2/3}); $\ell^2 R$ does not, and it enters only through the classical Λ term (P2).
3. **The size of every correction (deduced, with qualification Q2′).** Against the Einstein term, every correction is then a function F(x, $\ell^2 R$) with F(0, $\ell^2 R$) = 0 (P2). Where $\ell^2 R \gg 1$, P3 makes F tend to a function of x alone. Taking each term as a positive power of x, a correction is small when x is.
4. **The boundary (P4).** With no coefficient chosen, the boundary is x = 1: below it the corrections are guaranteed small; above it nothing here guarantees them small. Whether they are actually large there depends on coefficients P4 declines to fix. ∎

What P4 adds. The 1. Nature's own order-one factor, the unknown coefficients of the higher-curvature terms, is not derived here, and none is chosen.

Four dimensions (computed, this paper). With {G, ħ, c, R} the matrix again has rank 3, leaving one group, $l_P^2 R$. The same argument puts the boundary at $l_P^2 R = 1$.

What P3 excludes (computed). If a term $l_5 \ell R$ were admitted, growing with ℓ against P3, it would reach one at $R^* = 1/(l_5 \ell)$. There $x = l_5/\ell$, far before x = 1 whenever $\ell \gg l_5$. (The group $l_5 \sqrt{R^*} = (l_5/\ell)^{1/2}$ at the same point is not x; qualification Q1′.)

Relation to Haggard and Rovelli's criterion (READ; relation computed, this paper). Haggard and Rovelli (arXiv:1407.0989, eqs. (11)–(13), p. 4) argue "on dimensional grounds" that a process of duration τ stays classical only while $q = l_P R \tau \ll 1$. They also say: "Since this heuristic argument is not very strong, we leave b undetermined". Their criterion adds the elapsed time to the local curvature.

Evaluated on the local curvature time $\tau = R^{-1/2}$, it becomes $q = (l_P^2 R)^{1/2}$. So, at that time scale, their q is small exactly where the local group $l_P^2 R$ is, and the two criteria place the same onset up to an order-one factor that depends on the background (qualification Q3′). Haggard and Rovelli's bounce time $\tau \sim m^2/l_P$ (eq. (2), p. 2, READ) is the long-time effect their q adds. Proposition 8 does not need it.

What Proposition 8 does not prove.

- P1–P3 are premises of effective field theory, not theorems.
- "Each term a positive power of x" is an assumption about the expansion's form (Q2′).
- A bound on K is not a bound on every curvature component, because the Lorentzian K is not positive-definite (Q4′).
- Which invariant measures the scale moves the boundary by an order-one factor.

12.1 What this says about the relation

Theorems 1–2 and Proposition 8 together give the relation the paper set out to state:

- **Where $x \ll 1$:** the law is Einstein's, up to corrections that vanish as $x \to 0$. Under Postulate O its one possible quantum correction at second order is zero, and quantum matter and information live inside the one light cone (Lemma 5).
- **Where $x \to 1$:** the classical law stops being guaranteed, and nothing proved here says what replaces it. Postulate O says only that whatever replaces it keeps the one cone.

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13 · Verification

The computations were re-derived twice before this paper, each time by a separate AI session in this project working read-only and re-deriving with its own code. These are in-project readings, not outside review.

First verification: Lemmas 6–7 and Theorem 2. Confirmed, with qualifications; no sub-claim refuted. It re-derived:

- the Boulware–Deser solution at $\kappa = 1$ and 0 (with 2α and $-\alpha$ failing);
- the three cones from the explicit E_{ab} in a temporal gauge;
- $c_T = 3 - 2A$, $c_V = A$ and $c_S = 2A - 1$, with the sum rule;
- the tensor cone by direct linearisation;
- the plane-wave ω against Reall, Tanahashi and Way (4.11)–(4.12);
- the anti-de Sitter wave's ω divided by $1 - 4\alpha k^2$, exactly;
- the scalar–Gauss–Bonnet equations (as in §11);
- c_T^2 and c_S^2 on the cosmology, and $\omega = \pm 8\sqrt{2}\, f'\varepsilon$ analytically on the wave.

It also re-read the quotations of the outside sources. Its qualifications, each carried at its claim above:

- **Q1.** $\alpha = 0$ follows under Postulate O, not from Theorem 1 alone; Theorem 1's cone statement is flat-space only (§9).
- **Q2.** CEMZ constrain $|\alpha| \gg l_5^2$ only, at tree level (§10.3).
- **Q3.** The divisor of Lemma 6 (d) vanishes at $8\alpha k_{CD}^2 = 1$. "For either sign" holds within the existence range (§10.2).
- **Q4.** The split is proved on the families computed and is generic per RTW, not proved wherever the Weyl curvature is nonzero (§10.3).
- **Q5.** Lemma 7 needs $V(\varphi_0) = V'(\varphi_0) = 0$ (§11).
- **Q6.** "Either sign of f'" in cosmology is a reflection; the sign of $c_T^2 - 1$ is set by $\ddot{f}$ and $\dot{f}$ (§11).
- **Q7.** The cosmology check is blind to the Weyl part, and the verifier's own checks close the gap (§11).
- **Q8.** The symmetry in Lemma 7 (a) is ω's, not ξ_0^2's (§11).

Second verification: Proposition 8. Confirmed, with qualifications; one computed value refuted. It re-derived the rank and null space of Buckingham's matrix and read Haggard and Rovelli in full. Qualifications carried in §12:

- **Q1′.** The counterfactual onset is x = l_5/ℓ; the value $(l_5/\ell)^{1/2}$ is a different group, and was refuted as x.
- **Q2′.** The form of the expansion is an assumption.
- **Q3′.** Haggard and Rovelli's criterion is q = l_P R τ, which adds a time; compared at the local curvature time it orders the same onset as x, up to a background-dependent factor.
- **Q4′.** A bound on K is not a bound on all curvature.

---

14 · What is not proved

- Anything beyond the experiments' precision.
- Postulate O. It is assumed, never derived.
- The cone splitting at every point of nonzero Weyl curvature. It is proved on the families computed and generic per RTW.
- α = 0 below $|\alpha| \sim l_5^2$ by scattering alone. Theorem 2 needs Postulate O there.
- The scalar–Gauss–Bonnet exclusion where $V(\varphi_0) \neq 0$ or $V'(\varphi_0) \neq 0$.
- The causal behaviour of theories beyond second order (Einsteinian cubic gravity, f(R)), and of theories with an infinite tower of higher spins.
- The form of the correction expansion, nature's order-one factor at x = 1, and what law replaces Einstein's there.
- Any theory of quantum gravity, and any unification. The result is one shared causal structure.

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15 · References

Each was re-read at source for this paper (alphaXiv PDF text, 2026-10-16), with page numbers as printed, except where marked "named, not re-read".

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    2022. 157, doi:10.1007/JHEP08(2022)157; arXiv:2205.08551v2. Eq. (1) p. 1; §2 p. 2; eqs. (12)–(13) p. 5.
12. H. M. Haggard and C. Rovelli, *Black hole fireworks: quantum-gravity effects outside the horizon spark black to white hole tunneling*, Phys. Rev. D 92 (2015) 104020, doi:10.1103/PhysRevD.92.104020; arXiv:1407.0989v2. Eq. (2) p. 2; eqs. (11)–(13) p. 4.
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Journal references and DOIs were checked against each paper's arXiv record on 2026-10-16; the arXiv versions above are the texts read.

Appendix A · Every computation, with its module and check

The computations are carried by four modules, imported by `qr_paper_check.py`, each with its own selftest and mutants: M1 (special and general relativity), M2 (the Gauss–Bonnet cones), M3 (scalar–Gauss–Bonnet) and M4 (the quantum onset).

claim	where	module	check
E^2 = p^2 c^2 + m^2 c^4, E = pc at m = 0	§4	M1	L1
boost invariants of every kind	§5	M1	L2
geodesics independent of m; bending 1.751 arcsec	§6	M1	L3 ×2
p = ⌊(n − 1)/2⌋: p(4) = 1, p(5) = 2	§7	M1	L4
ρ_B is unchanged under any local operation of Alice	§8	this paper	L5
Einstein’s law: every characteristic front is null	§10	M2	L6
three cones on a static background	§10	M2	L6
c_T = 3 − 2A, c_V = A, c_S = 2A − 1; sum rule; excess 16|α|m/(r⁴q²)	§10	M2 + this paper	L6 ×2
ω = 8αε, 4αε, −8αε; sum zero; ÷ (1 − 4αk²) on anti-de Sitter	§10	M2	L6 ×2
sGB field equations, symbol	§11	M3	I3
(1 + 8f̈)/(1 + 8Hf̈), Reall’s (91)	§11	M3	L7
537/461; 8|Hf̈| = 1/25; 76/461; 1363/11525	§11	M3 + this paper	L7 ×2
|ξ|²/(1 ∓ 8√2 f̈ε)	§11	M3	L7
rank 3, two groups; in that basis ħ^{2/3} in l_5^2R only	§12	M4 (S1–S2)	B1, I4
four dimensions: one group l_P² R	§12	this paper	B1
x = l_5/ℓ at the counterfactual onset	§12	this paper	B2
q = (l_P² R)^{1/2} at τ = R^{−1/2}	§12	this paper	B3

Module suites, re-run for this paper (Appendix C):

- M1: 6/6 selftest, 4/4 mutants;
- M2: 10/10, 5/5;
- M3: 7/7, 3/3;
- M4: S1–S2 asserted directly by the check (its S5 asserts the superseded value (l_5/ℓ)^{1/2} and is not used; see Q1′).

Appendix B · Source map

claim	rests on
Lemma 1	computed (L1); READ Peres–Terno pp. 1, 13
Lemma 2	computed (L2); READ GW170817 eq. (1), Will abstract
Lemma 3	READ Will p. 6, Deser pp. 1, 4; computed (L3); READ MICROSCOPE eq. (5), GW170817 eq. (4) and p. 7
Lemma 4	READ Navarro–Navarro Thm 2.6, p. 1; computed count (L4); READ Charmousis–Dufaux pp. 3, 5, 16
Lemma 5	READ Peres–Terno p. 13; computed (L5)
Theorem 1	deduced from Lemmas 1–5; qualification Q1
Lemma 6	computed (L6, M2); READ RTW pp. 2, 13, 14, 17, 22, 24–25
Theorem 2	deduced from Lemma 6 + Postulate O + (PL); READ Charmousis–Dufaux p. 10; corroborated, READ CEMZ, within Q2
Lemma 7	computed (L7, M3); READ Reall eq. (91), SSTT eqs. (1), (12)–(13); Q5–Q8
Proposition 8	computed (B1–B3); premises P1–P4; READ Haggard–Rovelli eqs. (2), (11)–(13); Q1′–Q4′
§13	the two in-project AI re-derivations

Appendix C · Reproduction

```
pip install sympy                     # sympy 1.14.0 used
export PYTHONDONTWRITEBYTECODE=1
python3 qr_paper_check.py --selftest
python3 qr_paper_check.py --mutants
```

The check runs the own checks of M1–M3 and asserts M4's Buckingham steps (S1–S2) directly, before it asserts anything of its own. Run for this version:

- `--selftest`: 23/23 claims, M1–M3's own checks passing (I1–I3) and M4's S1–S2 holding (I4);
- `--mutants`: 6/6 caught, each by the claim it targets (`paper_drift` by L7, `signal` by L5, `onset_sqrt` by B2, `inst_hbar_in_y` by I4, `sum_weights` by L6, `inst_gb_double` by I2 and L6).

Tools. The text was drafted, and the sources re-read, with Claude (Anthropic), an AI assistant, working with the author. Quotations were read from the arXiv PDFs through alphaXiv; journal references through Firecrawl. Key results (Tangherlini's Kretschmann scalar, the cone sum rule, the scalar–Gauss–Bonnet numbers, light bending,

Buckingham’s ranks) were recomputed independently in Wolfram Language. The reviews mentioned in §13 were separate AI sessions in this project; none is an outside review.